temperature (≤40°C) multiply and convert milk into curd. Curd is more nutritious than milk as it contains a number of organic acids and vitamins.



Prebiotics are compounds in food (fibers) that induce the growth or activity of beneficial microorganisms.

Probiotics are live microorganisms intended to provide health benefits when consumed, generally by improving or restoring the gut flora.

Yogurt is produced by bacterial fermentation of milk, and lactic acid is produced as a byproduct. Microorganisms such as *Streptococcus thermophilus* and *Lactobacillus bulgaricus* coagulate the milk protein and convert the lactose in the milk to lactic acid. The flavour in yogurt is due to acetaldehyde.

Cheese is a dairy product produced in a wide range of flavours, textures and is formed by coagulation of the milk protein, casein. During cheese production, milk is usually acidified and the enzyme rennet is added to cause coagulation. The solids are separated and pressed to form cheese. Most cheese are made with a starter bacteria, *Lactococcus*, *Lactobacillus* or *Streptococcus*.

Paneer (cottage cheese) is fresh cheese common in South Asia, especially in India. It is made by curdling milk with lemon juice, vinegar and other edible acids. Large holes in Swiss cheese is due to the production of large amount of carbon-di-oxide by the bacterium *Propionibacterium shermanii*.

The dough used in the preparation of idlis and dosas are fermented by the bacteria *Leuconostoc mesenteroides* whereas the dough used in bread making is fermented by *Saccharomyces cerevisiae* (Baker's Yeast). Fermentation of glucose mainly forms ethyl alcohol and carbon-di-oxide, which is responsible for leavening of dough. When leavened dough is baked, both carbon-di-oxide and ethyl alcohol evaporate making the bread porous and soft.

Single cell protein (SCP)

Single cell protein refers to edible unicellular microorganisms like *Spirulina*. Protein extracts from pure or mixed cultures of algae, yeasts, fungi or bacteria may be used as ingredient or as a substitute for protein rich foods and is suitable for human consumption or as animal feed.

9.2 Microbes in industrial products

Microbes are used to synthesize a number of products valuable to human beings. Products like beverages, antibiotics, organic acids, amino acids, vitamins, biofuels, single cell protein, enzymes, steroids, vaccines, pharmaceutical drugs, etc., are produced in industries. Production on a large scale requires growing microbes in very large vessels called fermentors. A fermentor (bioreactor) is a closed vessel with adequate arrangement for aeration, agitation, temperature, pH control and drain or overflow vent to remove the waste biomass of cultured microorganisms along-with their products.

9.2.1 Antibiotic production

Antibiotics are chemical substances produced by microorganisms which can kill or retard the growth of other disease causing microbes even in low concentration. Antibiotic means "**against life**". Antibiotics are used to treat diseases such as plague, meningitis, diphtheria, syphilis, leprosy, tuberculosis etc., **Selman Waksman** discovered Streptomycin and was the first to use the term "**antibiotic**" in 1943.

While working on *Staphylococci* bacteria, Alexander Fleming observed a green mould growing in one of his unwashed culture plates around which *Staphylococci* could not grow. He

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